NITHISH KUMAR V

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PROFESSIONAL SUMMARY

MCA graduate (CGPA 8.97/10) with hands-on experience building Python/FastAPI applications and integrating Google Gemini. Built AI-assisted finance workflows and a document-Q&A RAG prototype (embeddings, retrieval, citation validation), and implemented machine-learning research with PyTorch and scikit-learn. Seeking to build Generative AI features for digital learning platforms.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript
AI / GenAI: Google Gemini API, LLM integration, prompt engineering, NLP, RAG, embeddings, pgvector
Backend & Data: FastAPI, REST APIs, SQLAlchemy, PostgreSQL, MySQL, Supabase
Machine Learning: PyTorch, scikit-learn, NumPy, LSTM autoencoders, Isolation Forest
Frontend & Tools: React, Next.js, Git/GitHub, Docker, Vitest

SELECTED PROJECTS

Finora — Family Finance Tracker | *React, Python, FastAPI, SQLAlchemy, PostgreSQL/Supabase, Gemini API* **2026**

- Built a personal-finance PWA with expense tracking, budgets, asset tracking, milestones, reports, and dashboards on a FastAPI and PostgreSQL/Supabase backend.
- Integrated the Google Gemini API through FastAPI for a conversational assistant, SMS-to-expense parsing, and AI-generated monthly insights.
- Fixed unreliable expense saving by requiring server acknowledgement, preserving drafts, warning on failure, and preventing duplicates; verified with 189 frontend and 95 backend tests and a passing production build (local).

Evidence-Grounded RAG — Local Prototype | *React, FastAPI, PostgreSQL/pgvector, embeddings* **2026**

- Built a document Q&A prototype for synthetic PDF/TXT files with text extraction, configurable chunking, embedding generation, and similarity retrieval.
- Added source-citation validation so answers trace back to retrieved passages, plus server-verified demo sessions and document-ownership checks.
- Covered the retrieval and Q&A workflow with 77 backend and 10 frontend tests, run offline without live external services.

Faculty Workload Management System (FWMS) | *Python, FastAPI, PostgreSQL, React, Docker, Power BI* **2025 – 2026**

- Developed a faculty workload management system with a PostgreSQL-backed data model, REST APIs, workload reporting, Excel exports, and role-based access controls.
- Implemented a five-pass allocation fallback strategy that progressively relaxes constraints to improve allocation feasibility.
- Built workload analytics against an 18-hour reporting threshold, Excel exports, and a three-page Power BI dashboard.

Chit Fund Analytics | *Next.js, TypeScript, React, Supabase, Vitest, GitHub Actions* **2026**

- Built a login-protected application to replace WhatsApp and spreadsheet chit-fund records, with a rule-based parser for raw WhatsApp auction messages and round tracking.
- Protected data integrity with an append-only ledger using idempotency keys against duplicate payouts, row-level security for per-user isolation, a verification workflow, and decimal-based (Big.js) ROI eligibility checks.
- Added Excel and printable statements plus audit exports; verified with 498 passing Vitest tests (17 placeholders skipped) and GitHub Actions CI running tests, type-checking, and the build.

RESEARCH

Nithish Kumar V (first author) and Nathiya R, "Shift-Aware Threshold Adaptation for Cross-Dataset Wearable Anomaly Detection" — IEEE ICSCST 2026, HITS Chennai.

- Implemented the academic study in Python (PyTorch, scikit-learn): LSTM autoencoder/CNN–LSTM feature extraction, Isolation Forest, and KS/KL-divergence-based threshold adaptation across the WISDM and UCI HAR datasets.

EDUCATION

MCA — Hindustan Institute of Technology & Science, Chennai — CGPA 8.97/10 · 2026
BCA — Hindustan College of Arts & Science, Chennai — CGPA 8.20/10 · 2024
HSC — Atomic Energy Higher Secondary School-2, Kalpakkam · State Board — 85.86% · 2021
SSC — Atomic Energy Central School-2, Kalpakkam · CBSE — 80.40% · 2019

CERTIFICATIONS & ACTIVITIES

NPTEL — Introduction to Algorithms & Analysis; Cloud Computing
Intellithon'25, 24-Hr National Hackathon, HITS
Practical Data Analytics using SQL & Cloud Services Workshop, HITS